

LA HW

Included exercises: 7, 13, 22, 36, 39-44

Exercises 1-8

In Exercises 1–8, find a basis for (a) the column space and (b) the null space of each matrix.

Original Exercise 7

$$\begin{bmatrix} -1 & 1 & 2 & 2 \\ 2 & 0 & -5 & 3 \\ 1 & -1 & -1 & -1 \\ 0 & 1 & -2 & 2 \end{bmatrix}$$

Exercises 9-16

In Exercises 9–16, a linear transformation T is given. (a) Find a basis for the range of T . (b) If the null space of T is nonzero, find a basis for the null space of T .

Original Exercise 13

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 + 2x_3 - x_4 \\ 2x_1 + x_2 + x_3 \\ 0 \\ 3x_1 + x_2 + x_4 \end{bmatrix}$$

Exercises 17-32

In Exercises 17–32, find a basis for each subspace.

Original Exercise 22

$$\left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathcal{R}^3 : \begin{array}{l} -x_1 + x_2 + 2x_3 = 0 \text{ and} \\ 2x_1 - 3x_2 + 4x_3 = 0 \end{array} \right\}$$

Exercises 33-52

In Exercises 33–52, determine whether the statements are true or false.

Original Exercise 36

If $\mathcal{S}$ is a linearly independent set and $\text{Span } \mathcal{S} = V$, then $\mathcal{S}$ is a basis for V .

Original Exercise 39

Every basis for a particular subspace contains the same

Original Exercise 40

The columns of any matrix form a basis for its column

Original Exercise 41

The pivot columns of the reduced row echelon form of A

Original Exercise 42

The vectors in the vector form of the general solution of

Original Exercise 43

If V is a subspace of dimension k , then every generating

Original Exercise 44

If V is a subspace of dimension k , then every generating